

CS351 - IT Workshop II (Cloud Computing)

Assignment 5

Total Marks: 40

Please follow the instructions below. Marks will be deducted if you do not follow these instructions.

(i) Screenshots: Take clear screenshots, resize the images to reduce their file size, and upload them to Google Classroom. Please do not upload a .zip file in Google Classroom.

(ii) Source Code File Naming: Write different Python files for different questions. Name each Python source file using the following format:
<assignment_no>_<question_no>.py

For Example: a1_q1.py

(iii) Source Code Header: Include your name and roll number at the top of each source file using the following format.

Name: <Your Name>

Roll No: <Your Roll No>

(iv) Proper Comments: You need to write comments for each block of code, function inputs and outputs, and the working logic of the functions.

(v) Proper Variable and Function Names: Please use meaningful variable and function names.

Part 1: Your Website and Startup Script

[10 Marks]

You are required to ***set up and host a new website*** on an AWS EC2 instance.

- Create the website with at least two HTML pages and CSS/images.
- Write a startup script that:
 - Installs and starts the Apache HTTP server.
 - Copies the website files to the Apache document root (/var/www/html).

- Displays the EC2 instance ID on the homepage (using instance metadata).

Part 2: Automation using boto3

[10 Marks]

Automate the **Website (Part 1)** Deployment using Python and boto3.

- Write a Python script that:
 - Creates a Security Group that allows inbound HTTP traffic on port 80.
 - Launches an EC2 instance using:
 - The startup script from Part 1.
 - A specified AMI (Ubuntu LTS).
 - An t3.micro instance type.
 - Attaches the Security Group to the instance.
- After launching, retrieve and print:
 - The instance ID.
 - The instance Public DNS name.
- Test your script by opening the public DNS in a browser and confirming the website works.

Part 3: Auto Scaling with CloudWatch

[20 Marks]

Extend your automation to create an **Auto Scaling Group (ASG)** with **CloudWatch alarms** and **scaling policies**.

- **Modify your Python script to:**
 - Create a **Launch Template** or **Launch Configuration** that uses:
 - The same AMI, instance type, Security Group, and startup script *from previous parts*.
 - Create an **Auto Scaling Group** in an Availability Zone (i.e. Mumbai (ap-south-1)) with:
 - Min size: 1, Max size: 3, Desired capacity: 1.
 - Configure **scale-out** and **scale-in** policies:

- **Scale Up:** If *average CPU $\geq 20\%$ for 2 minutes* → add 1 instance.
 - **Scale Down:** If *average CPU $\leq 30\%$ for 2 minutes* → remove 1 instance.
 - Create **CloudWatch alarms** to call these policies.
- **Test** auto-scaling by:
 - **Applying load** (e.g., *Apache Benchmark* or **stress**) to trigger *scale-up*.
 - **Removing load** to trigger *scale-down*.
- **Verify** by checking:
 - **Multiple EC2 instances** serving the site (confirm by refreshing and checking the *instance ID* on the page).
 - **CloudWatch alarms** transitioning between states.